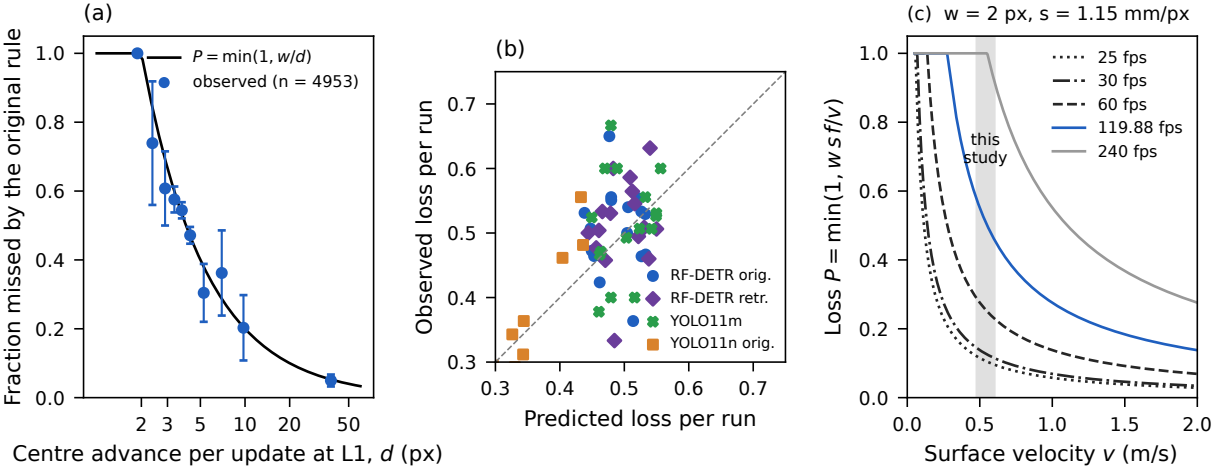
(a)
$P = \min(1, w/d)$
observed (n = 4953)
Fraction missed by the original rule
Centre advance per update at L1, d (px)
(b)
Observed loss per run
Predicted loss per run
RF-DETR orig.
RF-DETR retr.
YOLO11m
YOLO11n orig.
(c) w = 2 px, s = 1.15 mm/px
25 fps
30 fps
60 fps
119.88 fps
240 fps
this study
Loss $P = \min(1, w\,s\,f/v)$
Surface velocity v (m/s)